

VANSH KUMAR GARG

Software Engineer — Backend / Full-Stack Developer

 Portfolio —  GitHub —  LinkedIn

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Backend-focused Software Engineer with hands-on experience in building and debugging scalable APIs and database-driven systems. Strong foundation in SQL, PostgreSQL, and backend workflows, with practical experience in handling concurrency, optimizing queries, and resolving production-level issues. Comfortable working with logs, debugging API failures, and improving system reliability.

EDUCATION

Thapar Institute of Engineering and Technology, Patiala, Punjab

B.E. in Computer Science and Business Systems

Sep 2022 – May 2026

CGPA: 7.89

EXPERIENCE

Web Developer Intern — EllocentLabs

Feb 2026 – Present

- Designed and debugged backend APIs using FastAPI for core business logic, validation, and error handling.
- Resolved API issues by analyzing request/response flows, improving endpoint stability.
- Wrote and optimized SQL queries in PostgreSQL (joins, aggregations), ensuring data consistency.
- Managed schema migrations using Alembic and ensured smooth database updates across environments.
- Collaborated with frontend teams to debug integration issues and fix API failures, improving end-to-end system reliability.
- Analyzed API failures using logs and request tracing to identify root causes and improve system performance.

PROJECTS

MovieTix – Online Movie Ticket Booking Platform

[Live Demo](#) — [GitHub](#)

- * **Tech:** Node.js, Express.js, MongoDB, Stripe, React.js, Tailwind CSS, JWT
- * Designed and implemented RESTful APIs for booking workflows using Node.js and MongoDB.
- * Solved race conditions in concurrent seat booking using atomic database operations, preventing double booking issues.
- * Debugged and optimized booking flow by identifying latency bottlenecks, improving API response time by 15%.
- * Structured database queries and indexing strategies to ensure efficient data retrieval under high concurrency.
- * Handled edge cases in real-time seat selection and ensured data consistency across multiple users.

Food-Delivery Website

[GitHub](#)

- * **Tech:** React.js, Redux Toolkit, Node.js, Express.js, MongoDB, Stripe, Postman, Tailwind CSS
- * Built a full-stack food delivery platform with role-based dashboards for admins and restaurants, supporting menu management and order monitoring.
- * Designed backend APIs for cart, order, and payment workflows, ensuring reliable data flow.
- * Managed the complete order lifecycle, including cart, order placement, and payment integration, ensuring secure transactions.
- * Debugged order and payment flow issues, identifying edge cases and reducing transaction failures.

SKILLS

Languages: C++, Python, JavaScript

Backend: Node.js, Express.js, FastAPI

Databases: MongoDB, MySQL, PostgreSQL

Frontend: React.js, Next.js

Tools: Git, Postman, Vercel, Stripe API, JWT

Database Skills: Query Optimization, Indexing, Data Modeling, Concurrency Handling

Debugging Tools: Postman, API Testing, Log Analysis (basic), Error Handling

ACHIEVEMENTS

- * Solved **600+** DSA problems on LeetCode (Rating: **1562**) across topics like DP, Trees, Graphs, and Arrays. [LeetCode](#)
- * Solved **200+** problems on GeeksforGeeks, focusing on algorithmic optimisation and time complexity.

RELEVANT COURSEWORK

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|--------------------------------|---------------------|---------------------------------|
| • Data Structures & Algorithms | • Computer Networks | • Software Design (Agile/Scrum) |
| • Operating Systems | • Database Systems | • Object-Oriented Programming |